

PANDUAN PEMBUATAN SMART GREENHOUSE IoT

ESP32-S3-N16R8 Soldered

Ikuti step-by-step dari atas ke bawah.

Jangan loncat step — setiap step bergantung pada step sebelumnya.

Kelompok 1 Pengantar Rekayasa Design

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FASE 0: Persiapan Software

Sebelum menyentuh hardware, siapkan software dulu.

Step 1: Download & Install Arduino IDE

1. Buka browser, pergi ke arduino.cc/en/software
2. Download Arduino IDE 2.x (pilih sesuai OS: Windows/Mac/Linux)
3. Install seperti biasa (Next > Next > Finish)
4. Buka Arduino IDE setelah install selesai

Step 2: Install ESP32 Board Support

Arduino IDE default hanya kenal Arduino Uno. Kita perlu menambahkan ESP32.

1. Buka Arduino IDE
2. File > Preferences
3. Di "Additional Board Manager URLs", paste:
https://espressif.github.io/arduino-esp32/package_esp32_index.json
4. OK > Tools > Board > Board Manager
5. Cari "esp32" > "esp32 by Espressif Systems" > Install
6. Tunggu selesai (5-10 menit)

Step 3: Install Library yang Dibutuhkan

Library = kode tambahan supaya ESP32 bisa berkomunikasi dengan sensor.



Tools > Manage Libraries, cari & install:

- a. "BH1750" by Christopher Laws
- b. "DHT sensor library" by Adafruit
- c. "Adafruit Unified Sensor" by Adafruit
- d. "ESP32Servo" by Kevin Harrington
- e. "ArduinoJson" by Benoit Blanchon

FASE 1: Rakit Hardware

Rakit rangkaian elektronik di breadboard.

PERINGATAN: Jangan colok adaptor/USB sampai semua wiring selesai dan dicek!

Step 4: Pasang ESP32 di Breadboard

Pasang ESP32-S3 di Breadboard 830 Point

Tancapkan di tengah di kedua sisi pin harus bisa diakses

ESP32-S3-DevKitC-1

ESPRESSIF

ESP32-S3 Specs
32-bit Xtensa® dual-core @240MHz
Wi-Fi IEEE 802.11 b/g/n 2.4GHz + BLE 5 Mesh
512 KB SRAM (16 KB SRAM in RTC)
384 KB ROM
45 GPIOs, 4x SPI, 3x UART, 2x I2C,
1x USB, 1x I2S, 1x DAC, 1x LED, 1x PWM, 1x ADC

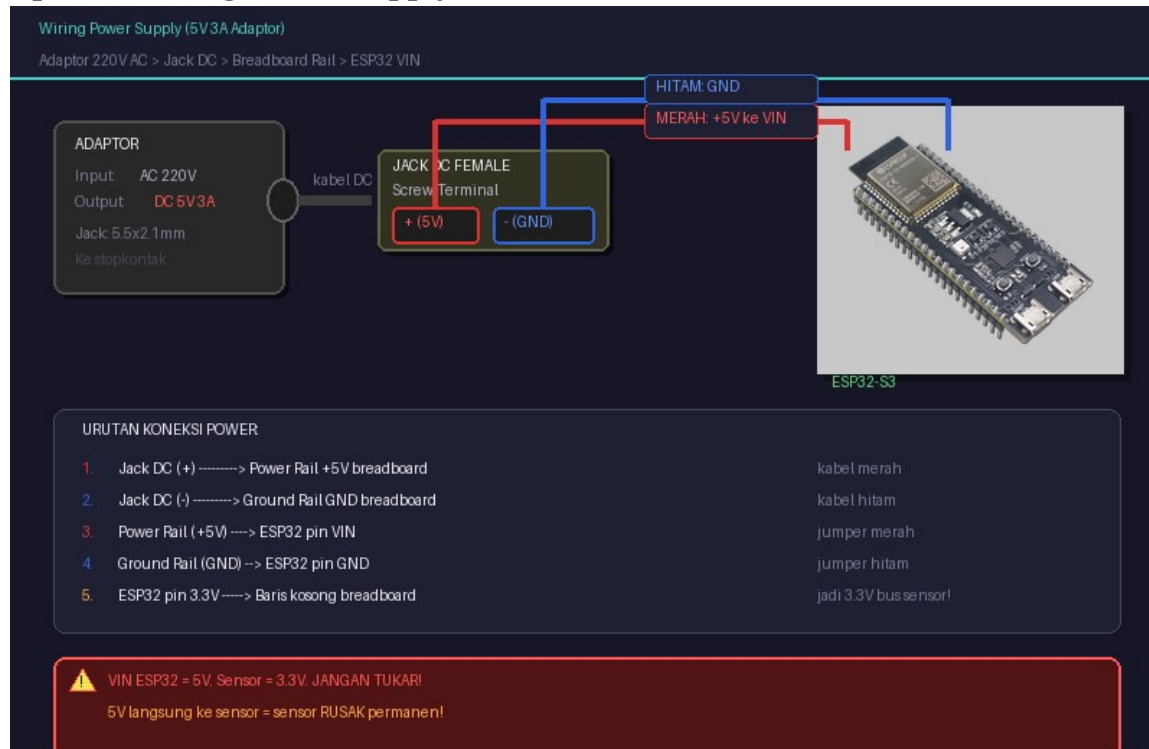
Legend:
PWM Capable Pin
GPIOX GPIO Input and Output
JTAG/USB JTAG for Debugging and USB
ADCK_CH Analog-to-Digital Converter
TOUCHX Touch Sensor Input Channel
OTHER Other Related Functions
SERIAL Serial for Debug/Programming
STRAP Strapping Pin Functions
RTC RTC Power Domain (VDD3P3_RTC)

Tekan ESP32 kuat semua pin masuk lubang. Kedua sisi harus bisa dipasangi jumper.

1. Ambil breadboard 830 point
2. Tancapkan ESP32-S3 di TENGAH breadboard
3. Kedua sisi pin harus bisa diakses untuk jumper wire

TIP: Tekan kuat tapi hati-hati. Semua pin harus masuk.

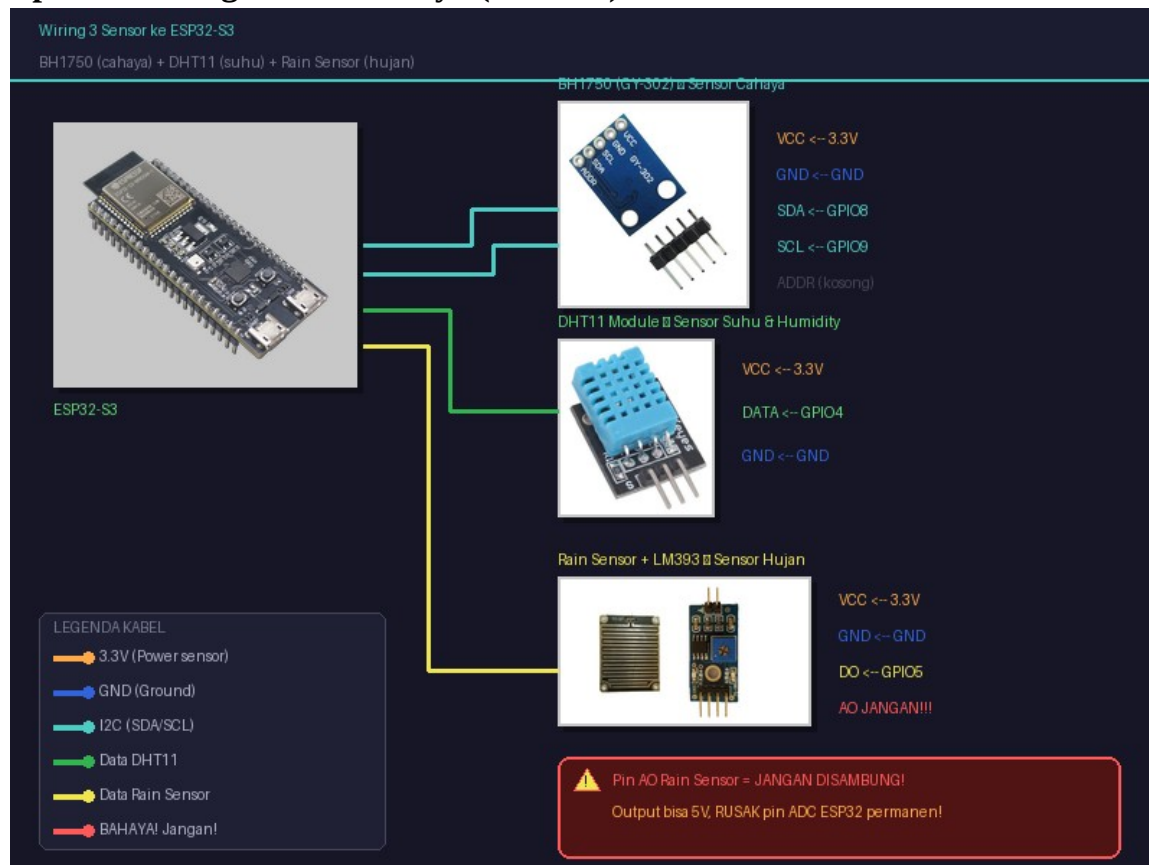
Step 5: Sambung Power Supply



1. Adaptor 5V 3A > jack DC female screw terminal (belum colok!)
2. **Terminal (+) > power rail (+) breadboard = +5V**
3. **Terminal (-) > ground rail (-) breadboard = GND**
4. Power rail (+) > ESP32 pin VIN
5. Ground rail (-) > ESP32 pin GND
6. ESP32 pin 3.3V > baris baru breadboard (jadi 3.3V bus sensor)

PERINGATAN: VIN ESP32 = 5V. Sensor = 3.3V. Jangan tukar!

Step 6: Sambung Sensor Cahaya (BH1750)



BH1750 pakai I2C (SDA + SCL):

VCC --> 3.3V bus

GND --> GND bus

SDA --> GPIO8

SCL --> GPIO9

ADDR --> kosong

Step 7: Sambung Sensor Suhu (DHT11)

DHT11 module 3-pin:

VCC --> 3.3V bus

GND --> GND bus

DATA --> GPIO4

TIP: Beli yang MODULE 3-pin (ada PCB), bukan komponen telanjang 4-pin.

Step 8: Sambung Sensor Hujan (Rain Sensor)

Dari control board LM393:

VCC --> 3.3V bus

GND --> GND bus

DO --> GPIO5

AO --> JANGAN DISAMBUNG!

PERINGATAN: Pin AO bisa output 5V. ESP32 ADC max 3.3V. Disambung = RUSAK PERMANEN.

Step 9: Sambung Servo Motor (MG996R)

Wiring Servo MG996R + Kapasitor 470uF
POWER SERVO = LANGSUNG DARI ADAPTOR 5V!



Servo MG996R
Torque: 11 kg.cm | 4.8-7.2V

Pinout MG996R (3 kabel warna)

Kabel MERAH (VCC)	+5V Power Rail (ADAPTOR!) Langsung dari adaptor, BUKAN ESP32!
Kabel COKLAT (GND)	GND Rail (ground bus) Ground bersama semua komponen
Kabel ORANYE (Signal)	GPIO13 ESP32 Sinyal PWM, kontrol posisi servo

⚠️ SERVO POWER = ADAPTOR 5V! BUKAN dari pin ESP32!
ESP32 max 500mA Servo stall = 2500mA Bisa RUSAK ESP32!

MERAH (VCC) --> +5V Rail (ADAPTOR!)

COKLAT (GND) --> GND bus

ORANYE (Signal) --> GPIO13

Capacitor 470uF dekat servo:

Kaki panjang (+) --> +5V Rail

Kaki pendek (-) --> GND Rail

PERINGATAN: Servo power WAJIB dari adaptor. BUKAN ESP32! Servo butuh 2.5A, ESP32 max 500mA.

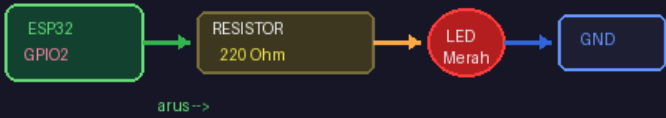
Step 10: Sambung LED Warning + Resistor

Wiring LED Warning + Resistor 220 Ohm
GPIO2 --> Resistor 220 Ohm --> LED Merah --> GND



LED Merah 5mm
Kaki panjang = + (Anoda)
Kaki pendek = - (Katoda)

RANGKAIAN SERI:



DETAIL:

Resistor 220 Ohm:	Tidak ada polaritas (boleh dibalik)
LED 5mm:	Kaki PANJANG = Anoda (+), Kaki PENDEK = Katoda (-)
Fungsi:	Berkedip saat Anti-Oven warning aktif (tutup + panas > 33C)
Kenapa 220 Ohm?	Membatasi arus ~15mA supaya LED tidak terbakar

GPIO2 > Resistor 220 Ohm > LED Merah > GND

Resistor: tidak ada polaritas. LED: kaki panjang = (+).

FASE 2: Upload & Test Firmware

Step 11: Checklist Sebelum Power On

CHECKLIST SEBELUM POWER ON
Cek SEMUA sebelum colok adaptor ke listrik!

☒

WAJIB Semua GND terhubung ke SATU ground bus

☒

WAJIB Tidak ada kabel 5V masuk langsung ke sensor (harus 3.3V!)

☒

WAJIB Rain sensor pin AO TIDAK tersambung ke manapun

☒

WAJIB Servo power dari ADAPTOR 5V, BUKAN dari pin ESP32

☒

WAJIB Capacitor 470uF polaritas benar: (+) ke 5V, (-) ke GND

☒

WAJIB Tidak ada kabel/komponen yang saling menyentuh (short)

☒

ESP32 belum di-power (belum colok adaptor/USB)

☒

Kabel USB yang disiapkan = kabel DATA (bukan charge-only)



Kalau ada yang SALAH = komponen RUSAK PERMANEN!
Double-check! Lebih baik lambat tapi aman.

Cek SEMUA item sebelum colok adaptor!

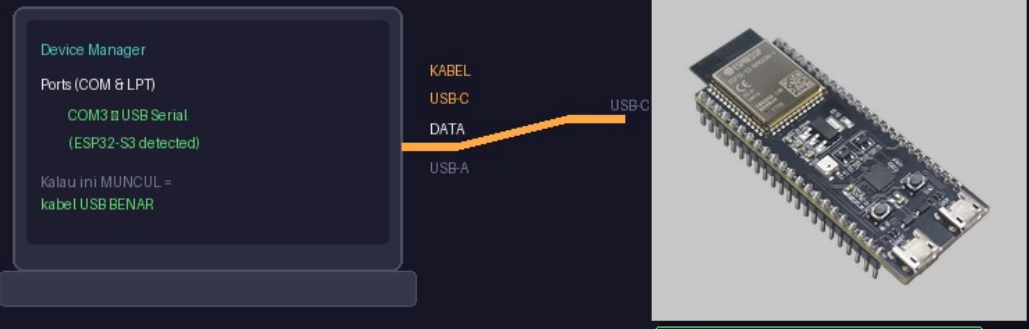
Setelah yakin:

1. Colok adaptor > cek LED power ESP32 menyala
2. Tunggu 10 detik > sentuh komponen
3. Kalau ada yang PANAS > CABUT SEGERA

Step 12: Colok ESP32 ke Laptop via USB

Colok ESP32-S3 ke Laptop via USB-C

Koneksi pertama: ESP32 ke laptop dengan kabel USB-C DATA



Device Manager

Ports (COM & LPT)

COM3 as USB Serial
(ESP32-S3 detected)

Kalau ini MUNCUL =
kabel USB BENAR

KABEL
USB-C
DATA
USB-A

ESP32-S3-N16R8 Soldered

⚠ Kabel HARUS USB DATA!
Kabel charge-only TIDAK punya jalur data D+/D-

Cara cek: Device Manager
Harus muncul COM port baru (COM3/COM4)

Data cable: 4 kabel internal (VCC, D+, D-, GND) as bisa transfer data
Charge-only: 2 kabel internal (VCC, GND saja) as TIDAK bisa transfer data

1. Kabel USB-C DATA (bukan charge-only!)
2. Colok ke ESP32 dan laptop
3. Device Manager harus muncul COM port baru

Tidak muncul? Ganti kabel USB atau install driver CH340/CP2102.

Step 13: Konfigurasi Arduino IDE

Setting Board di Arduino IDE

Tools > Board Settings as setiap setting yang merah = KRITIS

Arduino IDE 2.x as Tools > Board Configuration

Board: ESP32S3 Dev Module Board yang dipakai:

Board manager ESP32 by Espressif

PSRAM: OPI PSRAM

KRITIS! Harus OPI, BUKAN OSPI!

Flash Size: 16MB (128Mb)

Sesuai spek board N16R8

USB CDC On Boot: Enabled

KRITIS! Tanpa ini Serial Monitor MATI

Upload Speed: 921600

Speed tercepat

Partition Scheme: Default 4MB with spiffs

Default cukup

Port: COM3 (atau yg muncul)

⚠ PSRAM = OPI as USB CDC = Enabled WAJIB BENAR!
Salah setting: board tidak terdeteksi atau PSRAM out of memory

Board: ESP32S3 Dev Module

PSRAM: OPI PSRAM

Flash Size: 16MB (128Mb)

USB CDC On Boot: Enabled

Upload Speed: 921600

Port: COM yang muncul

PERINGATAN: PSRAM = OPI & USB CDC = Enabled WAJIB BENAR!

Step 14: Siapkan File Firmware

1. Clone/download folder firmware/ dari repo
2. Buka smart_greenhouse.ino di Arduino IDE
3. Copy credentials.h.example > credentials.h
4. Edit credentials.h:

```
#define WIFI_SSID    "nama_wifi"
```

```
#define WIFI_PASSWORD "password_wifi"
```

```
#define SUPABASE_URL  "https://xxx.supabase.co"
```

```
#define SUPABASE_KEY  "eyJ..."
```

TIP: Isi WiFi dulu. Supabase nanti. Sistem tetap jalan offline.

Step 15: Upload Firmware ke ESP32

Upload Firmware ke ESP32-S3

Klik Upload > Compile > Upload > Done!

- 1 Buka smart_greenhouse.ino
File > Open di Arduino IDE
- 2 Board = ESP32S3 Dev Module
Tools > Board
- 3 Port = COM yang muncul
Tools > Port
- 4 **Klik UPLOAD**
Ikon panah kanan (Ctrl+U)
- 5 'Compiling sketch...'
Tunggu 1-2 menit
- 6 'Uploading...'
Progress bar berjalan
- 7 **SELESAI!**
'Hard resetting via RTS pin...'

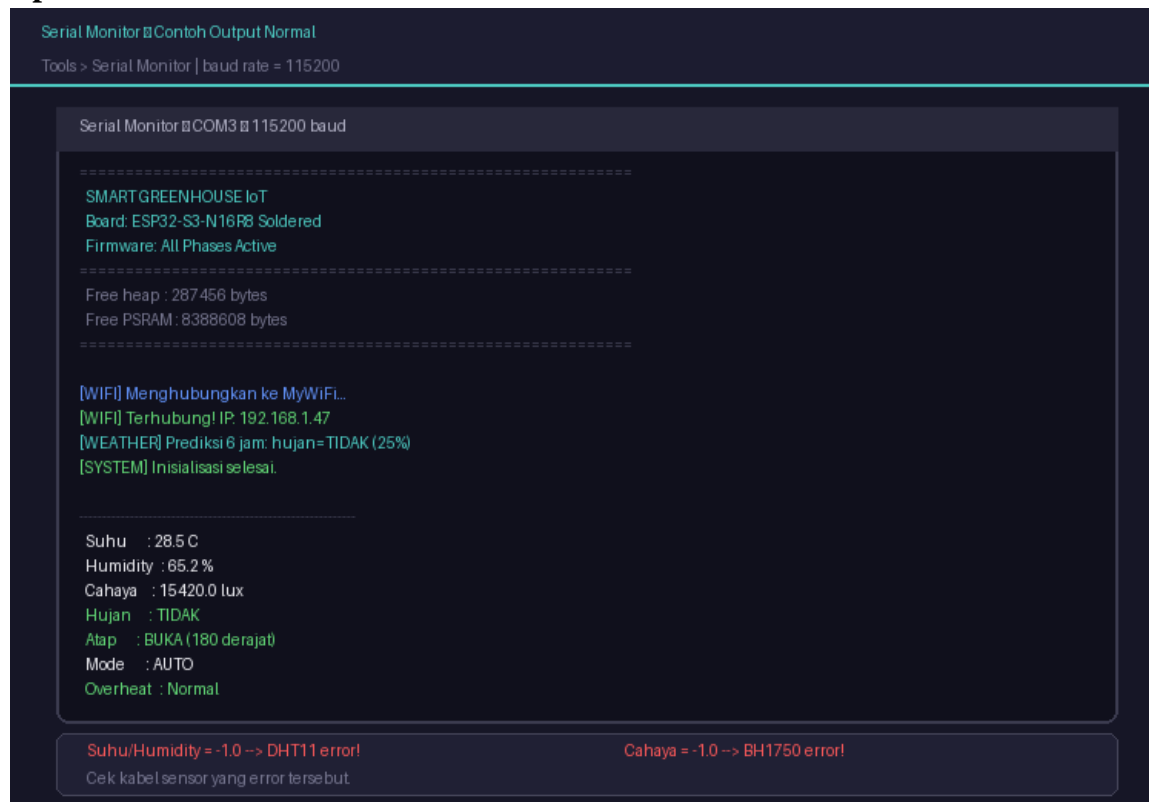
TROUBLESHOOT: Gagal 'Connecting.....'

1. Tahan tombol BOOT di board ESP32
2. Sambil tahan BOOT, tekan RESET sekali
3. Lepas RESET (BOOT masih ditahan)
4. Klik Upload di Arduino IDE
5. Setelah 'Uploading...' muncul, lepas BOOT

Klik Upload > tunggu compile > tunggu upload.

Kalau gagal "Connecting...": tahan BOOT > tekan RESET > lepas > upload > lepas BOOT

Step 16: Buka Serial Monitor



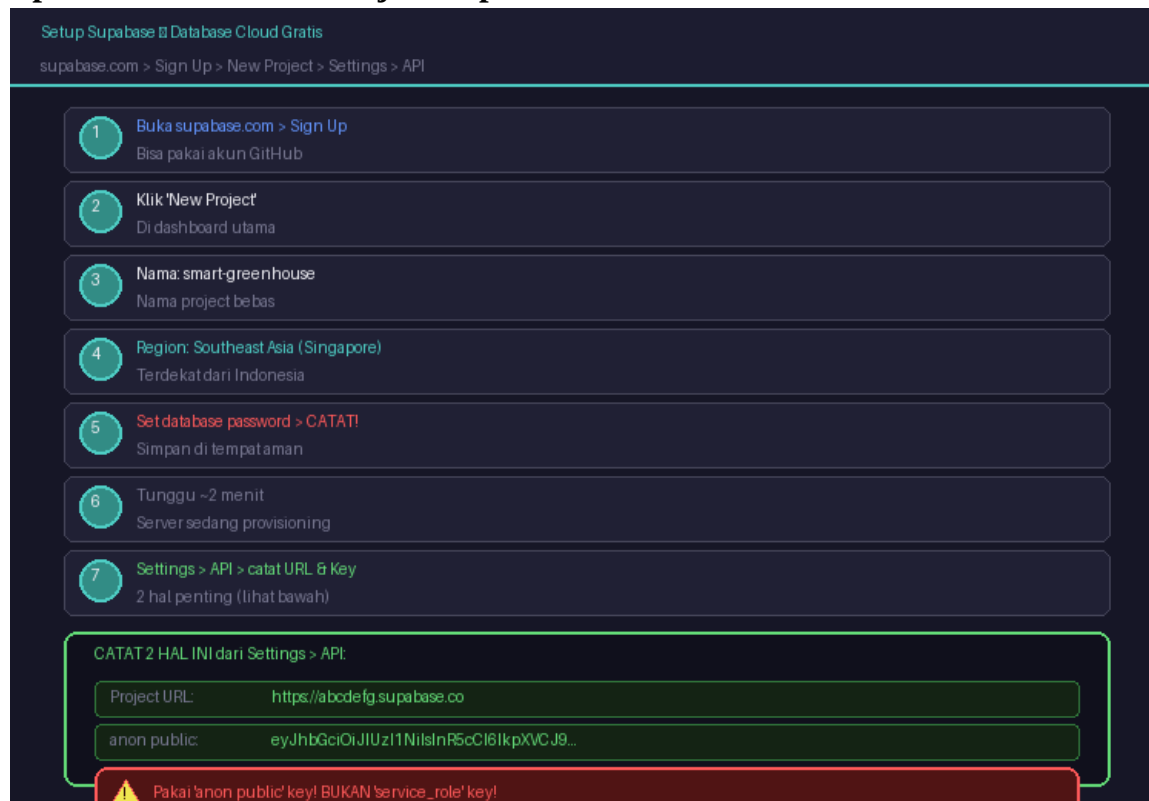
Tools > Serial Monitor, baud rate 115200.

Tekan RESET di ESP32. Harus muncul data sensor.

Kalau ada -1 = sensor error, cek kabelnya.

FASE 3: Setup Supabase

Step 17: Buat Akun & Project Supabase



supabase.com > Sign Up > New Project > smart-greenhouse

Region: Southeast Asia (Singapore)

Step 18: Catat URL dan API Key

Settings > API:

Project URL: `https://xxxxx.supabase.co`

anon public key: `eyJJ...`

Update credentials.h > upload ulang firmware.

PERINGATAN: Pakai "anon public" BUKAN "service_role" key!

Step 19: Buat Tabel Database

SQL Editor > New Query > paste & Run:

```
CREATE TABLE sensor_logs (  
  id BIGSERIAL PRIMARY KEY, temperature FLOAT, humidity FLOAT,  
  lux FLOAT, is_raining BOOLEAN DEFAULT FALSE,  
  roof_angle INTEGER DEFAULT 0, roof_state TEXT DEFAULT 'closed',  
  mode TEXT DEFAULT 'auto', overheating BOOLEAN DEFAULT FALSE,  
  created_at TIMESTAMPTZ DEFAULT NOW()  
);  
CREATE TABLE commands (  
  id BIGSERIAL PRIMARY KEY, action TEXT NOT NULL,
```

```
executed BOOLEAN DEFAULT FALSE,  
created_at TIMESTAMPTZ DEFAULT NOW(), executed_at TIMESTAMPTZ  
);  
ALTER TABLE sensor_logs ENABLE ROW LEVEL SECURITY;  
ALTER TABLE commands ENABLE ROW LEVEL SECURITY;  
CREATE POLICY "read" ON sensor_logs FOR SELECT USING (true);  
CREATE POLICY "insert" ON sensor_logs FOR INSERT WITH CHECK (true);  
CREATE POLICY "all" ON commands FOR ALL USING (true) WITH CHECK (true);
```

Step 20: Verifikasi Data Masuk

Tunggu 60 detik > Table Editor > sensor_logs > harus ada data baru!

FASE 4: Test Sistem Lengkap

Step 21: Test: Sensor Baca Normal

Serial Monitor tiap 5 detik: suhu (20-35C), humidity (40-90%), cahaya (0-65000 lux).

Step 22: Test: Hujan = Atap Tutup (Rule 1)



Teteskan air ke rain plate > "Hujan: YA" > tunggu 60 detik > servo tutup (0 derajat).

Rule 1: Local is King. Tidak bisa di-override.

Step 23: Test: Cerah = Atap Buka

Keringkan rain plate > arahkan senter ke BH1750 > tunggu 60 detik > servo buka (180).

Step 24: Test: Hysteresis

Tetes air > tunggu 30 detik > keringkan > servo TIDAK BOLEH bergerak!

Step 25: Test: Anti-Oven (Rule 4)

Hujan aktif + suhu > 33C: LED berkedip, atap tetap tutup.

FASE 5: Web Dashboard (Opsional)

Step 26: Setup Next.js

`npx create-next-app@latest greenhouse-dashboard`

`npm install @supabase/supabase-js`

`.env.local: NEXT_PUBLIC_SUPABASE_URL & NEXT_PUBLIC_SUPABASE_ANON_KEY`

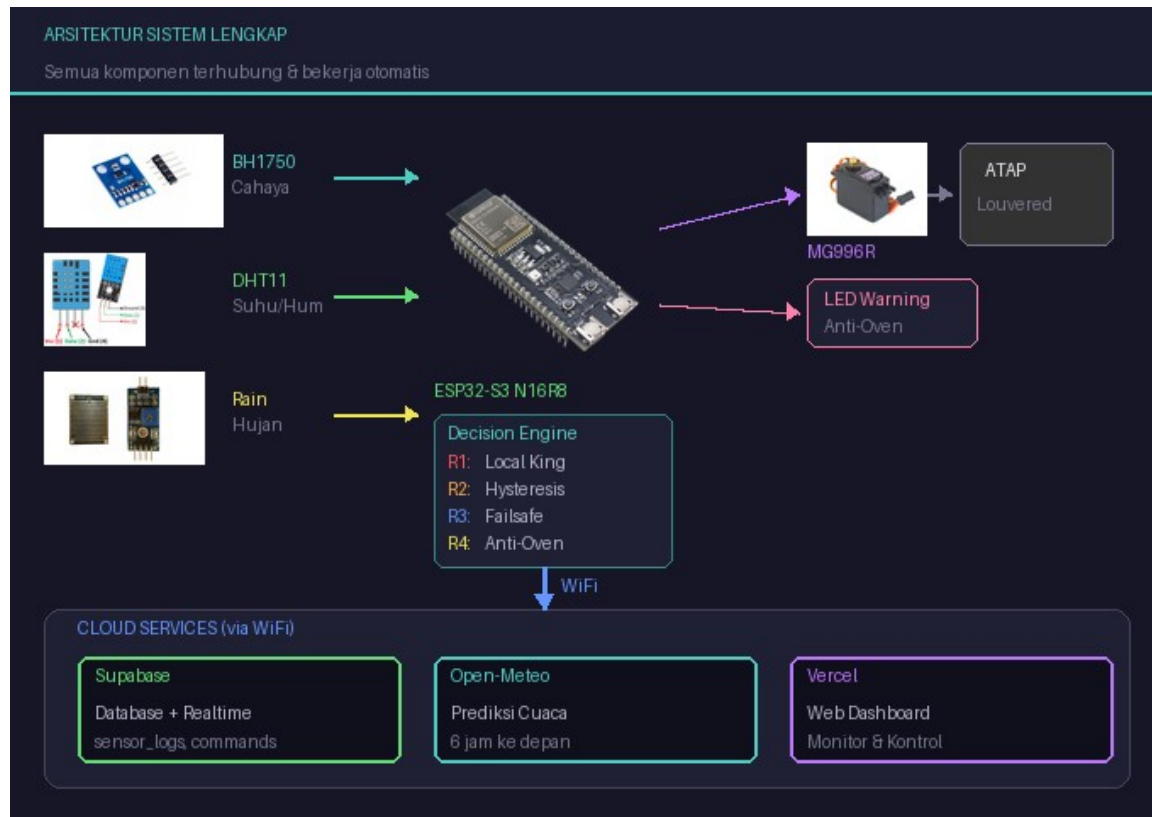
Step 27: Deploy ke Vercel

Push ke GitHub > vercel.com > Import > Add env vars > Deploy

Step 28: Test Dashboard

Data sensor muncul, tombol BUKA/TUTUP/AUTO berfungsi.

SISTEM LENGKAP



Selamat! Smart Greenhouse IoT sudah berjalan.

Sensor tiap 5s | Supabase tiap 60s | Command tiap 10s | Weather tiap 30min | Offline OK

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